

Memory statistics within the debug menu contain some unnecessary spaces

MC-252409 Debug

Status	Resolved / Fixed
Affected	1.18.2, 1.19 Pre-release 5, 1.19 Release Candidate 1, 1.19 Release Candidate 2, 1.19, 1.19.1 Pre-release 1 (+10 more)
Fixed in	23w51a
Reported	2022-06-02

STEPS TO REPRODUCE

1. Launch Minecraft 1.19 Release Candidate 1.
2. Create a new Creative world with default settings.
3. Enter the world.
4. Press F3 to open the debug screen.
5. Observe the memory statistics in the top-right area of the debug screen for unnecessary spaces.

OBSERVED BEHAVIOUR

The F3 debug screen displays extra whitespace in the memory statistics at the top right:

- The `Mem` and `Allocated` lines include an additional leading space before positive percentage values because their format string uses `% 2d`. This can result in two spaces between the label's colon and the percentage value, such as `Mem: 5%`.
- The `Allocation rate` line contains a space between the value and the unit separator, producing output such as `Allocation rate: 012MB /s`.

These spaces are formatting artifacts and are not needed for displaying the statistics.

EXPECTED BEHAVIOUR

The debug menu's memory statistics should be displayed without unnecessary spaces. The "Mem" and "Allocated" percentage values should use consistent spacing, and the "Allocation rate" value should not contain an extra space before "/s". All memory-related entries should remain clearly readable and correctly formatted while preserving the values shown.

ENVIRONMENT

Minecraft Java Edition Versions: 1.18.2, 1.19 Pre-release 5, 1.19 Release Candidate 1, 1.19 Release Candidate 2, 1.19, 1.19.1 Pre-release 1, 1.19.1 Pre-release 2, 1.19.1 Pre-release 5, 1.19.1 Release Candidate 2, 1.19.1, 1.19.2, 22w45a, 1.19.3, 1.19.4, 1.20.1, 1.20.3 Release Candidate 1 (includes snapshots)

ORIGINAL DESCRIPTION

The Bug:

Memory statistics within the debug menu contain some unnecessary spaces.

See MC-252409 - Analysis.png for all occurrences of this issue.

Steps to Reproduce:

- Enable the debug menu by hitting the "F3" key.
- Look towards the top right of the debug menu and look closely at the memory statistics.
- Take note as to whether or not memory statistics within the debug menu contain some unnecessary spaces.

Observed Behavior:

Unnecessary spaces are present.

Expected Behavior:

Unnecessary spaces would not be present.

Code Analysis:

Code analysis by Avoma can be found below.

The following is based on a decompiled version of Minecraft 1.19 Release Candidate 1 using Mojang mappings.

net.minecraft.client.gui.components.DebugScreenOverlay.java

```
public class DebugScreenOverlay extends GuiComponent {
```

```
...
```

```
protected List<String> getSystemInformation() {
```

```
...
```

```
ArrayList arrayList = Lists.newArrayList((Object[])new String[]{String.format("Java: %s %dbit",
System.getProperty("java.version"), this.minecraft.is64Bit() ? 64 : 32), String.format("Mem: % 2d%%
%03d/%03dMB", l4 * 100L / l, DebugScreenOverlay.bytesToMegabytes(l4),
DebugScreenOverlay.bytesToMegabytes(l)), String.format("Allocation rate: %03dMB /s",
DebugScreenOverlay.bytesToMegabytes(this.allocationRateCalculator.bytesAllocatedPerSecond(l4))),
String.format("Allocated: % 2d%% %03dMB", l2 * 100L / l, DebugScreenOverlay.bytesToMegabytes(l2)), "",
String.format("CPU: %s", GlUtil.getCpuInfo()), "", String.format("Display: %dx%d (%s)",
Minecraft.getInstance().getWindow().getWidth(), Minecraft.getInstance().getWindow().getHeight(),
GlUtil.getVendor()), GlUtil.getRenderer(), GlUtil.getOpenGLVersion()});
```

```
...
```

If we look at the above class, we can see that memory statistics within the debug menu contain some unnecessary spaces. This is evident through the following pieces of code:

```
... String.format("Mem: % 2d%% %03d/%03dMB" ...
```

```
... String.format("Allocation rate: %03dMB /s" ...
```

```
... String.format("Allocated: % 2d%% %03dMB" ...
```

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-252409>